Simulation Study Plan

shg_mode_survey · geometry × material design matrix, agreed 2026-10-12

This documents the campaign before any of it runs -- ten geometry/material combinations (MATRIX, next page), grouped into four underlying cross-section topologies (one page each, with a real diagram where the family already exists, a schematic where it doesn't yet).

Built from the generalized-geometry sketches shared 2026-10-12: a 3-layer stack (A = base film/rib, C = core ridge, B = loading/shell region), where whether B just caps C's top or wraps around its sides, and how large B is, span both the guided-mode families already in this repo (ridge, loaded, the corrected hybrid) and the not-yet-built Cerenkov case in one consistent vocabulary -- not four unrelated structures.

READING THE MATRIX

status: ✓ ready to run now, with existing materials data.
○ blocked on a materials gap (d33(Sc), GaAlN n/k/d33, or an anisotropic-substrate decision) -- see the Input Assumptions report for what's known and what isn't.
h_A: the base-layer thickness, free 0 (pure ridge/no embedding base) up to a full film (rib/loaded) -- only wired up in the `loaded` family today; adding it to the embedded topologies (rows 6-9) is new geometry work, not yet done (marked TBD*).

Not a commitment to run all ten at once -- user direction, 2026-10-12: check in before each batch.

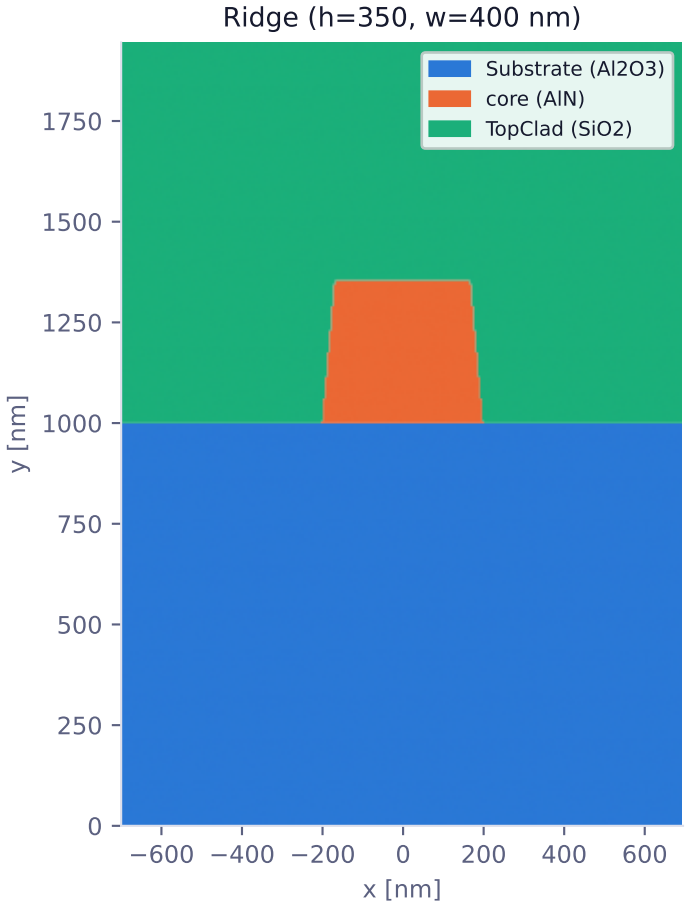
Design matrix

10 rows, grouped by topology — see the following pages for diagrams of each

#		TOPOLOGY	H_A	CORE (C)	SHELL/OUTER (B)	SUBSTRATE	PURPOSE	STATUS
1	✓	Ridge	n/a (no A layer)	AlN	—	Al2O3	baseline	done (`ridge`)
2	○	Ridge	n/a	ScAlN	—	Al2O3	does Sc-enhanced d33 alone beat AlN, no loading tricks	need d33(Sc)
3	✓	Rib-loaded	0 → film thickness	AlN	TiO2	Al2O3	baseline	done — found bad (SH-absorbing)
4	✓	Rib-loaded	0 → film thickness	AlN	SiN	Al2O3	SiN strip instead of TiO2	ready, not yet run
5	○	Rib-loaded	0 → film thickness	AlN	ScAlN	Al2O3	nonlinear-active loading strip — new idea	need d33(Sc)
6	✓	Embedded (guided)	0 → TBD*	AlN	SiN	Al2O3	corrected `hybrid` — Honda et al. mechanism	ready, re-run pending
7	○	Embedded (guided)	0 → TBD*	AlN	ScAlN	Al2O3	both-active variant	need d33(Sc)
8	○	Embedded (Cerenkov)	0 → TBD*	ScAlN	AlN (blanket)	Al2O3 or AlN	buried high-index channel radiating into thick AlN	need d33(Sc)
9	○	Embedded (Cerenkov)	0 → TBD*	GaAlN	AlN (blanket)	Al2O3 or AlN	same, GaAlN channel	need full GaAlN n/k/d33
10	○	any (cross-cutting)	—	—	—	AlN instead of Al2O3	substrate swap	eqns exist, anisotropic treatment TBD

Topology 1: Ridge

real cross-section, from geometry.py



MATRIX ROWS

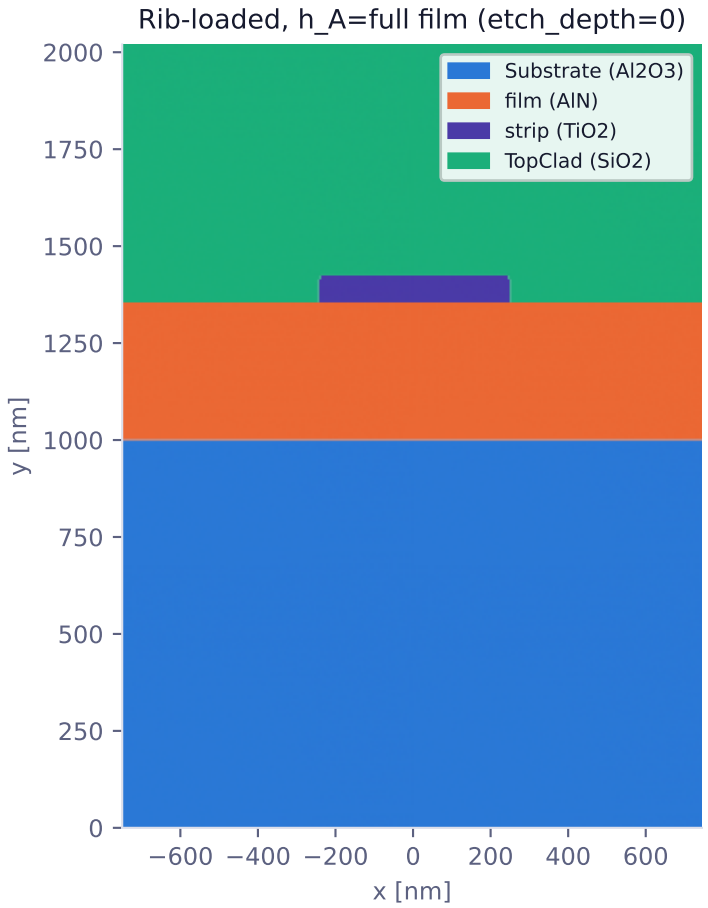
✓ **#1** AlN / — — baseline

○ **#2** ScAlN / — — does Sc-enhanced d33 alone beat AlN, no loading tricks

Bare AlN (or ScAlN) core directly on substrate -- no A layer, no B/shell at all. The simplest structure in the matrix, and the only one whose lossless NCE already survives real loss intact (ridge's own peak_efficiency is ~4-5 orders of magnitude better than every other family tried so far -- see the Input Assumptions / sub-metric reports). Row 2 asks whether Sc-doping the core alone (no shell, no loading) can beat this without needing any of the more complex topologies below -- blocked on the same d33(Sc) disagreement as rows 5/7/8.

Topology 2: Rib-loaded (cap-only)

real cross-section, from geometry.py



MATRIX ROWS

✓ **#3** AlN / TiO2 — baseline

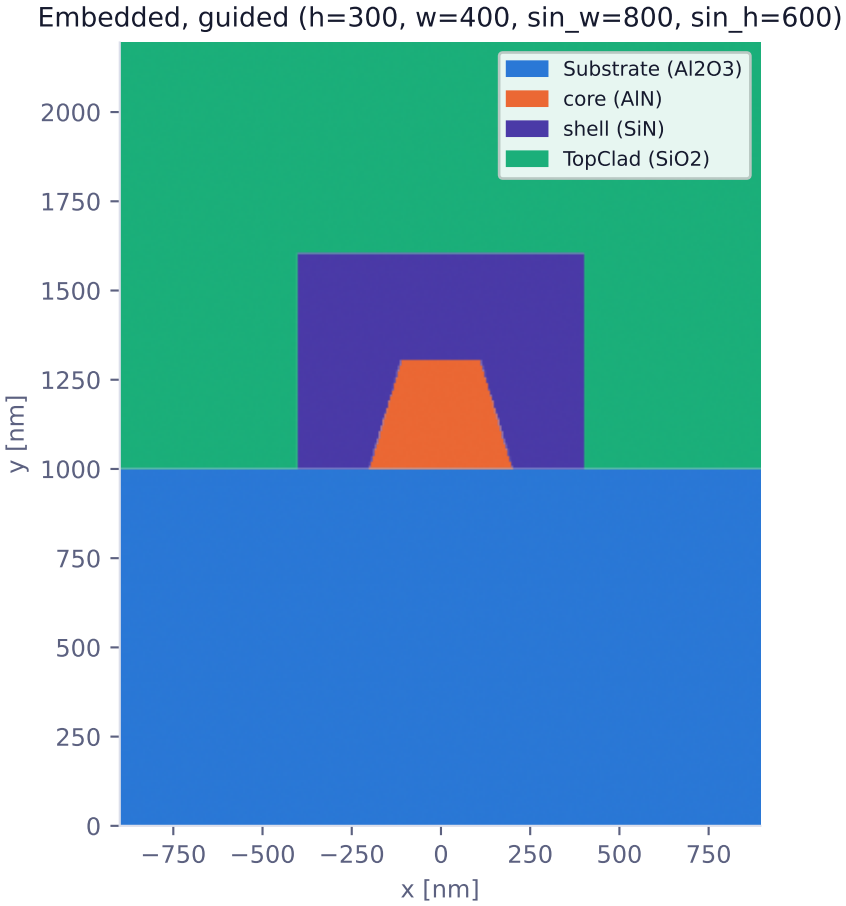
✓ **#4** AlN / SiN — SiN strip instead of TiO2

○ **#5** AlN / ScAlN — nonlinear-active loading strip — new idea

A (AlN film/rib) with a narrower loading strip B capping only its top (w_B <= w_C in the generalized sketch's terms) -- the existing `loaded` family. h_A ranges from a full, unetched film (shown here, etch_depth=0 -- the 'loaded waveguide' extreme) down to a fully etched ridge with just the strip's own confinement left. Row 3 (TiO2) already ran and was found bad -- TiO2 absorbs strongly at the SH wavelength, collapsing real efficiency ~40,000x below ridge's. Row 4 (SiN) swaps the strip material; row 5 asks whether making the strip itself nonlinear (ScAlN) changes the picture, pending the same d33(Sc) question as rows 2/7/8.

Topology 3: Embedded (guided)

real cross-section, from geometry.py



MATRIX ROWS

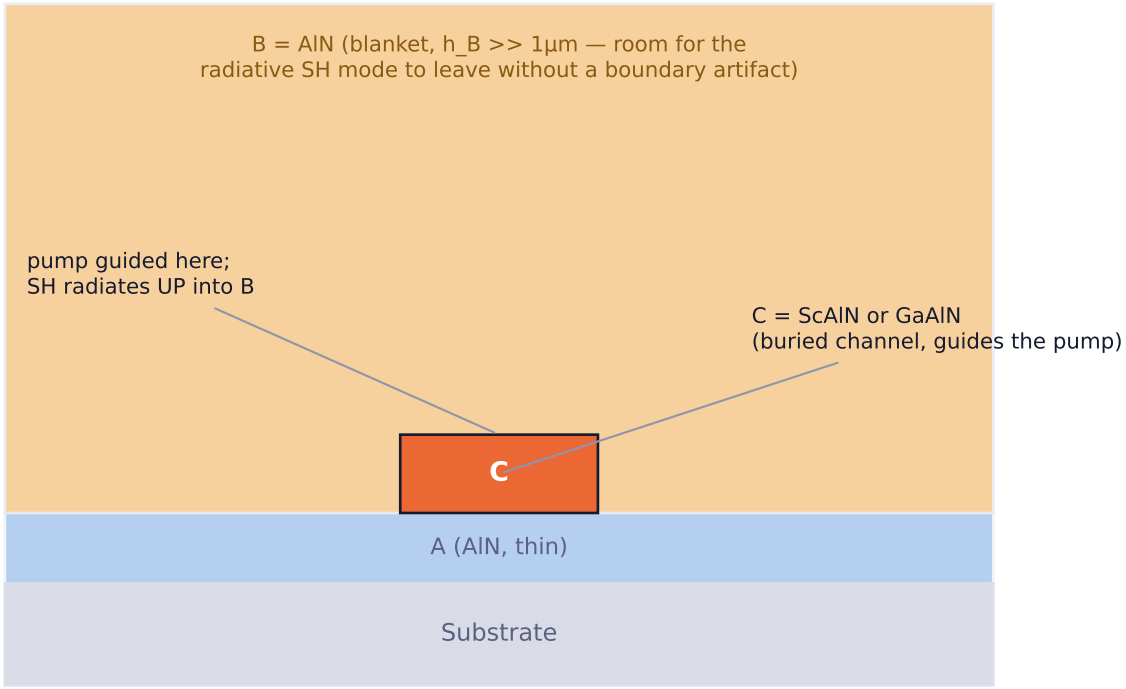
✓ **#6** AlN / SiN — corrected `hybrid` — Honda et al. mechanism

○ **#7** AlN / ScAlN — both-active variant

Core C wrapped on the sides (and above) by shell B (w_C <= w_B) -- the corrected `hybrid` family, fixed 2026-10-10 after the earlier 'pedestal' geometry bug (core was stacked on top of the shell, not embedded in it -- see PLAN.md). This is the structure that actually tests the Honda et al. 2025 TM00->TM20 mechanism; the re-run under the fix (row 6) is the next EMode campaign queued after this plan is reviewed. Row 7 swaps the shell for ScAlN, making both core and shell potentially nonlinear simultaneously -- pending d33(Sc).

Topology 4: Embedded (Cerenkov)

SCHEMATIC — not yet built



MATRIX ROWS

- ○ **#8** ScAlN / AlN (blanket) — buried high-index channel radiating into thick AlN
- ○ **#9** GaAlN / AlN (blanket) — same, GaAlN channel

Same wrap-around embedding as Topology 3, but the material roles flip and B grows large: C (small, buried) carries the enhanced index/nonlinearity (ScAlN or GaAlN) and guides the pump; B becomes a blanket AlN cladding thick enough (>>1 um) to host a genuinely radiative SH continuum rather than a tightly confined guided mode. No new plumbing needed for the Cerenkov physics itself -- `cerenkov_fdfd.py`'s driven 2D solve already takes eps_fn/d_fn/bbox generically from any geometry class, the same interface every existing family implements. What's missing is the geometry class itself (not yet written) and, for row 9 specifically, a live transparency check -- GaAlN's favorable (higher-d33) direction is Ga-rich, but GaN's bandgap sits below the SH photon energy, so the useful window may be narrow or may not exist; not yet checked quantitatively (Input Assumptions report, section 5).

Sequencing

Proposed order, each step its own check-in (user direction, 2026-10-12: not a green light to run all ten rows unsupervised) --

1. Row 6: re-run `hybrid1`/`hybrid_sweep1` (lateral) and `hybrid_vertical1` (vertical, extended thinner per the 'smaller sin_height keeps winning' trend from the pedestal data) under the corrected, now-embedded geometry. Already queued, blocked only on this plan's review.
2. Row 4: SiN rib-loaded, cheap (materials already wired), fills in topology 2 properly now that TiO2 (row 3) is known bad.
3. Rows 2/5/7/8: resolve the ScAlN d33 question first (sputtered vs. MBE disagreement, Input Assumptions report section 6) -- via the team's own ellipsometry data (repo pointer pending) and/or the Baumler 2019 0% Sc UV-k data point (PLAN.md, 2026-10-12). Rows 2/5/7 can then run with real numbers; row 8 additionally needs the embedded-Cerenkov geometry class (topology 4) built, which doesn't exist yet.
4. Row 9: GaAlN. Needs n(lambda)/k(lambda) sourced first (Input Assumptions report, section 5) -- lowest priority given the bandgap/transparency concern, unless that check turns up favorable.
5. Row 10: substrate swap, cross-cutting -- can run alongside any of the above once the anisotropic-vs-isotropic AlN-substrate treatment is settled (user direction, 2026-10-12: anisotropic).